Kollimarla Sai Sanjay

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Professional Summary

Software Engineer with 3+ years of experience designing and building scalable backend systems using Node.js, TypeScript, JavaScript, SQL, MongoDB, Redis and AWS. Experienced in architecting multi-tenant SaaS applications, developing distributed backend services, optimizing database performance, designing asynchronous processing pipelines, and building real-time applications. Strong foundation in Data Structures & Algorithms, System Design, Distributed Systems, Database Design, Scalability, Performance Optimization and end-to-end ownership of production software.

Technical Skills

Languages: Java, TypeScript, JavaScript, SQL, Python

Backend: Node.js, Express.js, REST APIs, BullMQ, Redis, JWT Authentication, WebSockets, Server-Sent Events (SSE)

Frontend: React.js, Angular, HTML, CSS, Bootstrap

Databases: MongoDB, MySQL, Redis

Cloud & DevOps: AWS S3, Docker, Git, Postman, Webpack

Core Concepts: Data Structures & Algorithms, System Design, Distributed Systems, Database Design, Data Modeling, Scalability, Fault Tolerance, High Availability, Caching, Concurrency, Asynchronous Processing, Event-Driven Architecture, Performance Optimization, Design Patterns, CI/CD, RBAC

Professional Experience

NexusIQ Solutions – Software Engineer | Feb 2024 – Present

- Designed and implemented a multi-tenant backend architecture serving multiple organizations with secure tenant isolation and configurable authorization.
- Architected RESTful backend services using Node.js and TypeScript supporting concurrent exam workflows with role-based access control across 40+ permission types.
- Optimized MongoDB queries through indexing, aggregation pipeline improvements and schema redesign, reducing API latency for high-volume operations.
- Built asynchronous processing pipelines for AI-assisted question generation and evaluation, improving content generation efficiency by over 70%.
- Integrated AWS S3 storage for scalable media management with secure upload/download workflows.
- Designed real-time browser monitoring services using Web APIs and WebSockets to support low-latency exam supervision.
- Owned features through architecture, implementation, deployment, production support and performance optimization.

Freelance – Full Stack Developer | May 2023 – Jan 2024

- Delivered end-to-end MEAN/MERN applications from requirements gathering through architecture, development and deployment for 5+ SME clients.
- Optimized API response times by up to 40% through database query optimization, indexing and caching.

Upgrad Insofe – Junior Software Engineer | Nov 2022 – Apr 2023

- Built Angular components and Node.js APIs for analytics dashboards, reducing page load time by 30% through lazy loading.
- Collaborated with data engineering teams to integrate machine learning models into production APIs.

Verzeo – Machine Learning Intern | May 2021 – Jun 2021

- Developed SVM-based diabetes predictor and handwritten digit recognizer achieving 98% accuracy on the MNIST dataset.
- Worked with Pandas, Scikit-learn and TensorFlow for model development.

Projects

Production-grade Email Queue Platform

Node.js • TypeScript • BullMQ • Redis • Docker • REST APIs • SSE • Webhooks

- Architected a distributed asynchronous email processing platform using BullMQ and Redis.
- Designed retries, exponential backoff, dead-letter queues (DLQ), idempotent processing and failure recovery.
- Built horizontally scalable worker architecture with delayed jobs, priority queues and rate limiting.
- Implemented real-time delivery tracking using SSE and webhook callbacks.
- Optimized Redis queue processing for high throughput and low latency.
- Containerized API and worker services using Docker.

Distributed Rate Limiting Service

Node.js • TypeScript • Redis • Lua Scripts • Express

- Designed a centralized distributed rate limiting service using Redis.
- Implemented Token Bucket and Sliding Window algorithms.
- Leveraged Redis atomic operations and Lua scripts to eliminate race conditions.
- Built reusable middleware for configurable request throttling.
- Benchmarked algorithms for latency, memory utilization and scalability.

TinyURL – Distributed URL Shortening Service

Node.js • TypeScript • MongoDB • Redis • Docker • AWS

- Designed a scalable URL shortening platform with analytics and expiration policies.
- Optimized redirect latency using Redis caching and MongoDB.
- Designed efficient indexes for read-heavy workloads.

- Implemented duplicate detection, click tracking and expiration scheduling.
- Documented architecture covering caching, scalability, load balancing and fault tolerance.

Education

Geethanjali College of Engineering and Technology
Bachelor of Technology (Computer Science & Engineering)
2019 – 2023 | CGPA: 7.8/10 | Hyderabad, Telangana

Certifications

- Full Stack Development (HTML, CSS, JavaScript, React, Node.js) — NxtWave
- Java Full Stack — Wipro TalentNext
- Machine Learning using Python — Verzeo
- Git, SQL, Command Line & Responsive Design — NxtWave